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The breakdown point of the median of means tournament

Daniela Rodriguez [a,*], Marina Valdora [b]

[a] Instituto de Cálculo, Facultad de Ciencias Exactas y Naturales, University of Buenos Aires and CONICET, Argentina
[b] Departamento de Matemática and Instituto de Cálculo, Facultad de Ciencias Exactas y Naturales, University of Buenos Aires, Argentina

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ABSTRACT

The median of means tournament estimator is known to have optimal properties since it achieves a sub-Gaussian performance for all random distributions with finite variance. In this work, we study the robustness of this estimator by computing its breakdown point.

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1. Introduction

Consider a random sample $\mathbf{X} = \{\mathbf{X}_1, \ldots, \mathbf{X}_n\}$, with $\mathbf{X}_i \in \mathbb{R}^d$ for $i = 1, \ldots, n$ and $E(\mathbf{X}_1) = \boldsymbol{\mu}$. The most popular estimator for $\boldsymbol{\mu}$ is the sample mean $\frac{1}{n}\sum_{i=1}^{n} \mathbf{X}_i$. The sample mean is known to have a near-optimal behaviour whenever the distribution of $\mathbf{X}_i$ has sufficiently light tails. However, when heavy tails are a concern, the sample mean is to be avoided as it may have a sub-optimal performance.

For random vectors with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, an estimator $\hat{\boldsymbol{\mu}}_n$ is said to be sub-Gaussian if, for some constant $L > 0$ and for every $\delta > 0$ and $n \in \mathbb{N}$,

$$P\left(\|\hat{\boldsymbol{\mu}}_n - \boldsymbol{\mu}\| \leq L\left(\sqrt{\frac{Tr(\boldsymbol{\Sigma})}{n}} + \sqrt{\frac{\lambda_{max}\log(2/\delta)}{n}}\right)\right) \geq 1 - \delta, \tag{1}$$

where λ_{max} denotes the largest eigenvalues of $\boldsymbol{\Sigma}$. If $\mathbf{X}_1 \ldots \mathbf{X}_n$ have a Gaussian distribution, the empirical mean has this property.

The median-of-means estimator has been proposed and studied by different authors (Nemirovsky and Yudin, 1983; Jerrum et al., 1986; Alon et al., 2002) and is defined in the following way. The data are partitioned into k groups of roughly equal size, the empirical mean is computed in each group, and finally the median of the obtained values is taken as the final estimate. As shown in Devroye et al. (2016), this simple estimator achieves a sub-Gaussian performance for all distributions with a finite variance when $d = 1$. When $d > 1$, the extension of the median of means estimator to the multivariate setting in a way that it achieves a sub-Gaussian behaviour, is not straightforward. The multivariate median of means estimators defined using the natural extensions of the median (coordinate-wise median or geometric median), are not sub-Gaussian. Lugosi and Mendelson (2019), introduced a new mean estimator, called median of means tournament (MOM), that achieves a sub-Gaussian performance for all distributions whose covariance matrix exists. Recently, Hopkins

* Corresponding author.
E-mail address: drodrig@dm.uba.ar (D. Rodriguez).

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(2018) studied an algorithm for its computation in polynomial time. The MOM estimator is defined in the following way. We start by partitioning the set $\{1, \ldots, n\}$ into k blocks $B_1, \ldots, B_k$ of size $m = \left[\frac{n}{k}\right]$. To simplify the presentation we assume that $n = km$ and therefore the size of all the blocks is m. All the arguments in the proofs apply if n is not divisible by k and the blocks have different sizes but a more complicated notation is needed in that case. Call $\overline{\mathbf{X}}_i$ the sample mean within block i and $\overline{\mathbf{X}}$ the mean of these sample means, which coincides with the sample mean since all the blocks have the same size. For each $\mathbf{a} \in \mathbb{R}^d$ define the set

$T_{\mathbf{a}}(\mathbf{X}) = \left\{\mathbf{z} \in \mathbb{R}^d : \exists J \subset \{1, \ldots, k\}, \#J > \frac{k}{2} \text{ such that } \left\|\overline{\mathbf{X}}_j - \mathbf{z}\right\| \leq \left\|\overline{\mathbf{X}}_j - a\right\| \forall j \in J\right\}$, where $\#J$ denotes the numbers of elements of J. The MOM estimator is defined by

$$\hat{\boldsymbol{\mu}}(\mathbf{X}) \in \operatorname{argmin}_{\mathbf{a}\in\mathbb{R}^d} \operatorname{radius} T_{\mathbf{a}}(\mathbf{X}),$$

where $\operatorname{radius} T_{\mathbf{a}}(\mathbf{X}) = \sup_{\mathbf{z}\in T_{\mathbf{a}}(\mathbf{X})} \|\mathbf{z} - \mathbf{a}\|$.

Meanwhile, a common pitfall in most classical procedures, namely those based on maximum likelihood or least squares, is that they are very sensitive to deviations from model assumptions. Robust statistics is focused on the development of methods that permit trustworthy inference, even in the presence of a small proportion of observations that deviate from the model assumptions. For symmetric univariate distributions, M-estimators are optimal in their relationship between robustness and efficiency. For asymmetric distributions, the usual robust methods assume that the observations have a distribution that belongs to a parametric family $\{F(x, \theta), \theta \in \Theta \subset \mathbb{R}\}$. These methods include Hampel's optimal estimators, M-estimators based on the probability integral transformations, proposed in Valdora (2014), methods for generalized linear models such as those proposed in Bianco et al. (2013) and Valdora and Yohai (2014), among others, and methods for multivariate observations such as those studied by Maronna (1976), Davies (1987) and Donoho (1982), among others. For further details on these estimators see Maronna et al. (2018).

One of the basic tools used to describe and measure robustness is the breakdown point. Intuitively, the breakdown point of an estimator is the proportion of incorrect observations an estimator can handle before giving a completely incorrect result. For instance the sample mean has a breakdown point of 0 and the median has a breakdown point of 0.5. The finite sample breakdown point is an approximation of the asymptotic breakdown point based on a finite sample of size n. Its precise definition is in the following section.

It is the aim of this paper to study the breakdown point of the MOM estimator defined in Lugosi and Mendelson (2019), adding a property to this estimator. Although the breakdown point of the MOM estimator is not the optimal 0.5 achieved by some highly robust estimators, we remark that it is distribution free, as opposed to many robust estimators that require the data to satisfy certain properties. The contribution of this paper is to give the exact value of the finite sample breakdown point of the MOM estimator. This will be useful in applications, since it tells us how many outliers the estimator can handle.

The rest of the paper is organized as follows. In Section 2, we give the definition of the breakdown point of location estimators and its exact value for the MOM estimator. Section 3 is dedicated to some simulation results in which we show that the empirical mean absolute error of the MOM estimator is bounded if the proportion of outliers is smaller than the breakdown point found in Section 2. Finally, in Section 4 we prove the theorem presented in Section 2.

2. Main results

The finite sample breakdown point of a location estimator is the fraction of the data that can be given arbitrary values without making the estimator go to infinity. We are interested in studying the behaviour of the MOM estimator when the sample is contaminated with s outliers. We formalize this idea by defining the set $\mathcal{X}_s = \{\mathbf{Y} : \#\mathbf{Y} = n \text{ and } \#\mathbf{X} \cap \mathbf{Y} = n - s\}$. This set represents all the possible samples that result from contaminating the original sample $\mathbf{X}$ with s outliers. Let $s^* = \max\{s \geq 0 : \hat{\boldsymbol{\mu}}(\mathbf{Y}) \text{ be bounded for all samples } \mathbf{Y} \in \mathcal{X}_s\}$. Then the breakdown point of the estimator $\hat{\boldsymbol{\mu}}$ is defined by $\epsilon^*(\hat{\boldsymbol{\mu}}, \mathbf{X}) = s^*/n$. For details on this robustness measure see, for example Huber and Ronchetti (2009) and Maronna et al. (2018).

Theorem 1. *Given random sample $\mathbf{X} = \{\mathbf{X}_1, \ldots, \mathbf{X}_n\}$ with mean $\boldsymbol{\mu}$, let $\hat{\boldsymbol{\mu}}(\mathbf{X})$ be the MOM estimator of $\boldsymbol{\mu}$ based on $\mathbf{X}$. Then the breakdown point of $\hat{\boldsymbol{\mu}}(\mathbf{X})$ is $[(k-1)/2]/n$, where k is the number of blocks.*

As pointed out by Lugosi and Mendelson (2019), k is a parameter of the estimator whose value depends on the desired confidence level. This dependence is given by $k = [200 \log(1/\delta)]$, where $1 - \delta$ is the desired confidence level. Note that larger values of k result in smaller values of δ and therefore MOM estimators with a larger breakdown point have a higher confidence level. Also note that, due to (1), larger values of k correspond to larger confidence regions.

3. Simulation experiments

In this section, we report the results of some Monte Carlo simulation experiments we performed for univariate distributions. The computation of the MOM estimator for multivariate observations is difficult and we do not know of a good method for its computation, see Lugosi and Mendelson (2019) and Hopkins (2018). For this reason we only considered the univariate case. We considered 1000 replications of independent samples of size $n = 100$, with three

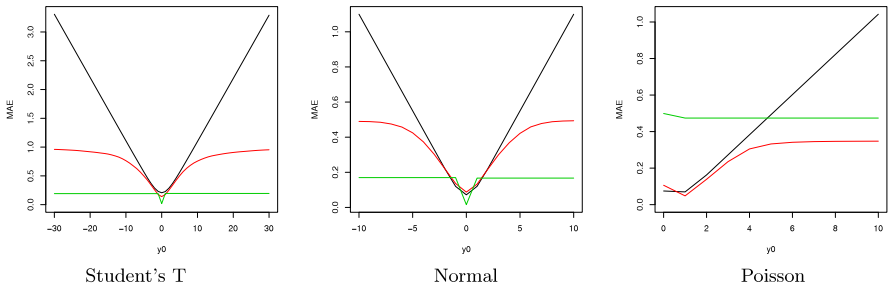

Fig. 1. Empirical MAE of estimators of the mean. Colours black, red and green correspond to the mean, the MOM and the median respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

different distributions: (a) Student's T with two degrees of freedom, (b) standard normal distribution and (c) Poisson with $\lambda = 1/2$. For the median of means estimator we chose the numbers of blocks $k = 8log(1/\delta)$, as suggested in Lugosi and Mendelson (2019) for univariate observations, taking $\delta = 0.05$. In each of the four scenarios, the samples were contaminated with $[bp.n]$ outliers, where $bp = [(k-1)/2]/n$. This means that $[bp.n]$ randomly chosen observations were replaced by identical outliers y_0, with y_0 ranging from -30 to 30 in case (a), from -10 to 10 in case (b) and from 0 to 10 in case (c). With the chosen values of δ and k, $bp = 0.11$ so the samples were contaminated with 11 outliers. In Fig. 1, we plot the empirical mean absolute error (MAE) as a function of the outlier y_0 for each estimation method. In this way, we show that the MAE of the MOM is bounded. For comparison, we also plot the MAE of the median and the mean which we know to have breakdown points of 0.5 and 0, respectively. As expected, the sample mean only has a good performance for samples without contamination, while the median has a good behaviour under symmetric distributions. However, as is well known, for asymmetric distributions the sample median is not a consistent estimator of the population mean, since it estimates a different parameter. In contrast, the MOM estimator performs well for symmetric and asymmetric distributions achieving a bounded mean absolute error. Based on these results, we conclude that the MOM estimator is a reasonable alternative when the proportion of outliers is expected to be small, specifically smaller than $k/(2n)$.

4. Proof of Theorem 1

To simplify notation, we assume k is odd.

Part 1 — We first prove that the breakdown point of the MOM estimator is not larger that $(k-1)/2n$ by showing that there exists a sequence of samples $\mathbf{Y}_N$ in $\mathcal{X}_{(k+1)/2}$ such that $\hat{\boldsymbol{\mu}}(\mathbf{Y}_N) \to \infty$ as $N \to \infty$.

Consider the sample in which the first $(k+1)/2$ elements are replaced by $N\mathbf{e}_1 = (N, 0, \ldots, 0)$. We will call these elements outliers. Suppose each of the outliers is in a different block; moreover assume they are in blocks $B_1, \ldots B_{(k+1)/2}$. We say these blocks are contaminated and so is the sample $\mathbf{Y}_N$ because they contain outliers. Let $\overline{\mathbf{Y}}_{N,i}$ be the mean of block i of the contaminated sample $\mathbf{Y}_N$ and $\overline{\mathbf{Y}}_N$ the mean of the $\overline{\mathbf{Y}}_{N,i}$ for $i = 1, \ldots, k$.

Step 1 — In this step we show that, if $\hat{\boldsymbol{\mu}}_N = \hat{\boldsymbol{\mu}}(\mathbf{Y}_N)$ is bounded, then radius $T_{\hat{\boldsymbol{\mu}}_N}(\mathbf{Y}_N) \to \infty$ as $N \to \infty$. Let $S = \sup_{N\in\mathbb{N}} \|\hat{\boldsymbol{\mu}}_N\|$.

Note that the set $T_{\hat{\boldsymbol{\mu}}_N}(\mathbf{Y}_N)$ can be written as

$T_{\hat{\boldsymbol{\mu}}_N}(\mathbf{Y}_N) = \left\{\mathbf{z} \in \mathbb{R}^d : \exists J \subset \{1, \ldots, k\}, \#J > \frac{k}{2} \text{ such that } z \in \overline{B}\left(\overline{\mathbf{Y}}_{N,j}, \|\overline{\mathbf{Y}}_{N,j} - \hat{\boldsymbol{\mu}}_N\|\right)\right\}$, where $\overline{B}(t, \epsilon)$ denotes the closed ball with centre t and radius ϵ. Thus, we write the set $T_{\hat{\boldsymbol{\mu}}_N}(\mathbf{Y}_N)$ in the following way

$$T_{\hat{\boldsymbol{\mu}}_N}(\mathbf{Y}_N) = \bigcup_{J\subset\{1\ldots k\}, \#J>k/2} \bigcap_{j\in J} \overline{B}\left(\overline{\mathbf{Y}}_{N,j}, \|\overline{\mathbf{Y}}_{N,j} - \hat{\boldsymbol{\mu}}_N\|\right).$$

Second, denote $\mathbf{X}_j^i$ the jth element in block i and assume the outliers are $\mathbf{X}_1^i$ for $1 \le i \le (k+1)/2$. Then,

$$\|\overline{\mathbf{Y}}_{N,i} - \overline{\mathbf{Y}}_{N,l}\| = \left\|\sum_{j=2}^{m} \frac{\mathbf{X}_j^i - \mathbf{X}_j^l}{m}\right\| \le \frac{m-1}{m} \max_{1\le i,j\le n} \|\mathbf{X}_i - \mathbf{X}_j\| = C \tag{2}$$

for all $1 \le i, l \le (k+1)/2$. Since $\hat{\boldsymbol{\mu}}_N$ is bounded and block i is contaminated, $\|\overline{\mathbf{Y}}_{N,i}\| \to \infty$ as $N \to \infty$ and then, if N is large enough, $\|\overline{\mathbf{Y}}_{N,i} - \hat{\boldsymbol{\mu}}_N\| > 2C$ for all $i \le \frac{k+1}{2}$. We will show that, if N is large enough, and $i \le (k+1)/2$, then $B\left(\overline{\mathbf{Y}}_{N,i}, \|\overline{\mathbf{Y}}_{N,i} - \hat{\boldsymbol{\mu}}_N\| - 2C\right) \subset \bigcap_{j=1}^{\frac{k+1}{2}} B\left(\overline{\mathbf{Y}}_{N,j}, \|\overline{\mathbf{Y}}_{N,j} - \hat{\boldsymbol{\mu}}_N\|\right)$. Let $i, j \le (k+1)/2$ and $\mathbf{x} \in B\left(\overline{\mathbf{Y}}_{N,i}, \|\overline{\mathbf{Y}}_{N,i} - \hat{\boldsymbol{\mu}}_N\| - 2C\right)$, using (2) we have

$$\begin{aligned}\|\mathbf{x} - \overline{\mathbf{Y}}_{N,j}\| &\le \|\mathbf{x} - \overline{\mathbf{Y}}_{N,i}\| + \|\overline{\mathbf{Y}}_{N,i} - \overline{\mathbf{Y}}_{N,j}\| \le \|\overline{\mathbf{Y}}_{N,i} - \hat{\boldsymbol{\mu}}_N\| - 2C + C \\ &\le \| - \overline{\mathbf{Y}}_{N,j}\| + \|\overline{\mathbf{Y}}_{N,j} - \hat{\boldsymbol{\mu}}_N\| - C \le \|\overline{\mathbf{Y}}_{N,j} - \hat{\boldsymbol{\mu}}_N\|.\end{aligned}$$

Now, for all $i \le (k+1)/2$, since

$$T_{\hat{\boldsymbol{\mu}}_N}(\mathbf{Y}_N) \supseteq \bigcap_{j=1}^{\frac{k+1}{2}} B\left(\overline{\mathbf{Y}}_{N,j}, \|\overline{\mathbf{Y}}_{N,j} - \hat{\boldsymbol{\mu}}_N\|\right) \supseteq B\left(\overline{\mathbf{Y}}_{N,i}, \|\overline{\mathbf{Y}}_{N,i} - \hat{\boldsymbol{\mu}}_N\| - 2C\right),$$

$$\operatorname{diam} T_{\hat{\boldsymbol{\mu}}_N}(\mathbf{Y}_N) \ge \operatorname{diam} B\left(\overline{\mathbf{Y}}_{N,i}, \|\overline{\mathbf{Y}}_{N,i} - \hat{\boldsymbol{\mu}}_N\| - 2C\right) = \|\overline{\mathbf{Y}}_{N,i} - \hat{\boldsymbol{\mu}}_N\| - 2C.$$

Therefore radius $T_{\hat{\boldsymbol{\mu}}_N}(\mathbf{Y}_N) \ge \frac{1}{2}\|\overline{\mathbf{Y}}_{N,i} - \hat{\boldsymbol{\mu}}_N\| - C$. The right hand side goes to ∞ as $N \to \infty$. This concludes Step 1.

Step 2 — In this step we prove that if we consider the vector $\tilde{\boldsymbol{\mu}}_N = N\mathbf{e}_1/m$, then the sequence $\{\text{radius } T_{\tilde{\boldsymbol{\mu}}_N}(\mathbf{Y}_N), N \in \mathbb{N}\}$ is bounded.

Without loss of generality, we assume that $\mathbf{x}_1 \in T_{\tilde{\boldsymbol{\mu}}_N}(\mathbf{Y}_N)$. Then there exists $J_1 \subset \{1, \ldots, k\}$ such that $\#J_1 > \frac{k}{2}$ and $\left\|\overline{\mathbf{Y}}_{N,j} - \mathbf{x}_1\right\| \le \left\|\overline{\mathbf{Y}}_{N,j} - \tilde{\boldsymbol{\mu}}_N\right\|$ for all $j \in J_1$. At least one of the blocks B_i with i in J_1 contains an outlier. Suppose it is B_{i_1}. Then $\|\mathbf{x}_1 - \tilde{\boldsymbol{\mu}}_N\| \le \left\|\overline{\mathbf{Y}}_{N,i_1} - \mathbf{x}_1\right\| + \left\|\overline{\mathbf{Y}}_{N,i_1} - \tilde{\boldsymbol{\mu}}_N\right\| \le 2\left\|\overline{\mathbf{Y}}_{N,i_1} - \tilde{\boldsymbol{\mu}}_N\right\|$. Note that $N\mathbf{e}_1$ is an element of B_{i_1} and denote with $\mathbf{X}_j^{i_1}$ for $2 \le j \le m$ the other $m-1$ elements, then

$$\overline{\mathbf{Y}}_{N,i_1} - \tilde{\boldsymbol{\mu}}_N = 1/m\left(\sum_{j=2}^{m} \mathbf{X}_j^{i_1} + N\mathbf{e}_1\right) - N\mathbf{e}_1/m = 1/m\sum_{j=2}^{m} \mathbf{X}_j^{i_1}$$

and therefore radius $T_{\tilde{\boldsymbol{\mu}}_N}(\mathbf{Y}_N) \le 2(m-1)/m \max_{1\le i\le n} \|\mathbf{X}_i\|$. This concludes Step 2.

Step 3 — We now show that Steps 1 and 2 imply that $\epsilon^*(\hat{\boldsymbol{\mu}}, \mathbf{X}) < \frac{k+1}{2n}$. We need to show that $\max \mathcal{S} < \frac{k+1}{2}$, where

$$\mathcal{S} = \{s \ge 0 : \hat{\boldsymbol{\mu}}(\mathbf{Y}) \text{ is bounded for all samples } \mathbf{Y} \in \mathcal{X}_s\}.$$

Let $s = (k+1)/2$. Consider the sequence of samples $\mathbf{Y}_N$ and suppose $\hat{\boldsymbol{\mu}}_N$ is bounded. Let $\tilde{\boldsymbol{\mu}}_N = N\mathbf{e}_1/m$, as defined in Step 2. We have proved that radius $T_{\hat{\boldsymbol{\mu}}_N}(\mathbf{Y}_N) \to \infty$ and that radius $T_{\tilde{\boldsymbol{\mu}}_N}(\mathbf{Y}_N)$ is bounded. Therefore, if N is large enough, radius $T_{\tilde{\boldsymbol{\mu}}_N}(\mathbf{Y}_N) <$ radius $T_{\hat{\boldsymbol{\mu}}_N}(\mathbf{Y}_N)$. This is a contradiction. We conclude $\hat{\boldsymbol{\mu}}_N$ is not bounded. Finally, note that, since $\mathcal{X}_{s+1} \supset X_s$, then if $s \notin \mathcal{S}$ neither is s', for all $s' \ge s$. Then, the fact that $(k+1)/2 \notin \mathcal{S}$ implies $\max \mathcal{S} < (k+1)/2$.

Part 2 — Now we prove that the breakdown point is at least $(k-1)/(2n)$. We will show that, if $s \le (k-1)/2$, then there exists M such that, for all $\mathbf{X} \in \mathcal{X}_s$, $\|\hat{\boldsymbol{\mu}}(\mathbf{X})\| < M$. This is done in 3 steps:

Step 1 — We show that for every fixed sample $\mathbf{Y} \in \mathcal{X}_s$, radius $T_{\mathbf{a}}(\mathbf{Y}) \to \infty$ as $\|\mathbf{a}\| \to \infty$. Since the number of outliers in the sample is not larger than $(k-1)/2$, there exists $J \subset \{1, 2, \ldots, k\}$ such that $\#J > k/2$ and all the blocks in J are free of outliers. For all $j \in J$, $\lim_{\mathbf{a}\to\infty} \|\overline{\mathbf{Y}}_j - \mathbf{a}\| = +\infty$. Then, for all M, there exists $\delta > 0$ such that $\|\mathbf{a}\| > \delta$ implies $\bigcap_{j\in J} B\left(\overline{\mathbf{Y}}_j, \|\overline{\mathbf{Y}}_j - \mathbf{a}\|\right) \supset B(\mathbf{0}, M/2)$, that in turn implies, diam $T_{\mathbf{a}}(\mathbf{Y}) >$ diam $\bigcap_{j\in J} B\left(\overline{\mathbf{Y}}_j, \|\overline{\mathbf{Y}}_j - \mathbf{a}\|\right) \ge M$ and, therefore, radius $T_{\mathbf{a}}(\mathbf{Y}) > M/2$.

Step 2 — We prove that, for every fixed sample $\mathbf{Y} \in \mathcal{X}_s$, if block j_0 is free of outliers, then radius $(T_{\overline{\mathbf{Y}}_{j_0}}(\mathbf{Y})) \le 2M_0$, where

$$M_0 = \max\left\{\|\overline{\mathbf{Y}}_{j_1} - \overline{\mathbf{Y}}_{j_2}\| \text{ such that } B_{j_1} \text{ and } B_{j_2} \text{ are free of outliers}\right\}.$$

Let $\mathbf{y}_1 \in T_{\overline{\mathbf{Y}}_{j_0}}(\mathbf{Y})$ then there exists J_1 with $\#J_1 > k/2$ such that $\mathbf{y}_1 \in \bigcap_{j\in J_1} \overline{B}\left(\overline{\mathbf{Y}}_j, \|\overline{\mathbf{Y}}_j - \overline{\mathbf{Y}}_{j_0}\|\right)$. Since the number of outliers is $< k/2$, at least one of the blocks in J_1 is free of outliers. Call it j_1. Note that either $j_1 \ne j_0$ or $\mathbf{y}_1 = \overline{\mathbf{Y}}_{j_0}$. Then

$$\|\mathbf{y}_1 - \overline{\mathbf{Y}}_{j_0}\| \le \|\mathbf{y}_1 - \overline{\mathbf{Y}}_{j_1}\| + \|\overline{\mathbf{Y}}_{j_1} - \overline{\mathbf{Y}}_{j_0}\| \le 2\|\overline{\mathbf{Y}}_{j_1} - \overline{\mathbf{Y}}_{j_0}\| \le 2M_0.$$

Step 3 — We combine Step 1 and Step 2 to get the result. Suppose that there exists a sequence of samples $\mathbf{Y}_N \in \mathcal{X}_s$, with $s < (k-1)/2$, such that $\|\hat{\boldsymbol{\mu}}_N\| = \|\hat{\boldsymbol{\mu}}(\mathbf{Y}_N)\| \to \infty$ as $N \to \infty$. Since $s < (k-1)/2$, there exists a block with index j_0 that has no outliers. We have proved that this implies that radius $T\hat{\boldsymbol{\mu}}_N(\mathbf{Y}_N) \to \infty$ as $N \to \infty$. On the other hand, since $\hat{\boldsymbol{\mu}}_N$ is a minimizer we have that, radius $T\hat{\boldsymbol{\mu}}_N(\mathbf{Y}_N) <$ radius $(T_{\overline{\mathbf{Y}}_{N,j_0}}(\mathbf{Y}_N)) < 2M_0$ for all N. This is a contradiction. We conclude that $\|\hat{\boldsymbol{\mu}}_N\|$ is bounded.

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